

Q4. Bitwise Operators

ANSWER

```

a=0b1101-binary-1101-13
b=0o25-octal-25-21
c=0x1A-hexadecimal-26
a&b(and)
a=13=01101
b=21=10101
13&21=00101-5
result
5=00101
26=11010
5|26=11111=31

bin(31)='0b11111'
oct(31)='0o37'
hex(31)='0x1f'
result>>2<31>>2=7
result<<1=31<<1=62
~result=-31=-32
DATA TYPES
a:int
b:int
c:int
result:int

```

Q5. binary search

CODE

```

def rotated_binary_search(arr,target):
    low=0
    high=len(arr)-1
    while low<=high:
        mid=(low+high)//2
        if arr[mid]==target:
            return mid
        if arr[low]<=arr[mid]:
            if arr[low]<=target<arr[mid]:
                high=mid-1
            else:
                low=mid+1
        else:
            if arr[mid]<target<=arr[high]:
                low=mid+1
            else:
                high=mid-1
    return -1
data=[4,5,6,7,0,1,2]
t=int(input("Target: "))
result=rotated_binary_search(data,t)
print("Found at: ",result)

```

INPUT

6

OUTPUT

Target: Found at: 2

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Q1. Harshad number

ANSWER

```
n=int(input("Enter a number:"))
temp=n
sum=0
while temp>0:
    digit=temp%10
    sum=sum+digit
    temp=temp//10
if n % sum==0:
    print("Harshad Number")
else:
    print("Not Harshad number")
print("Harshad number from 1 to 500:")
for i in range(1,501):
    temp=i
    sum=0
    while temp>0:
        digit=temp%10
        sum=sum+digit
        temp=temp//10
    if i % sum==0:
        print(i,end=" ")
```

Q2. GCD

CODE

```
a,b=map(int,input().split())
while b!=0:
    temp=a
    a=b
    b=temp%b
print("GCD is",a)
```

INPUT 6 3

OUTPUT {no output}

Q3. sum to a target value

ANSWER

```
def find_pairs(lst,target):
    seen=set()
    pairs=[]
    for num in lst:
        complement=target-num
        if complement in seen:
            pairs.append((complement,num))
        seen.add(num)
    return pairs
data=[int(x) for x in input("Enter numbers:").split()]
t=int(input("Target: "))
result=find_pairs(data,t)
print("pairs: ",result)
```